

DATABASE MANAGEMENT SYSTEMS (CS301)

Relational Model Concepts

Learning Objectives

Understand the relational model, relations, tuples, attributes, domains, keys, and referential integrity concepts.

Introduction to Relational Model

The Relational Model was proposed by E.F. Codd in 1970. It represents data in the form of tables called relations and is the most widely used database model today.

Relation

A relation is a table consisting of rows and columns. Each relation stores information about a particular entity or concept.

Tuple

A tuple represents a single row in a relation. Each tuple contains values for all attributes of the relation.

Attribute

An attribute is a column in a relation. It describes a property of an entity.

Domain

A domain is the set of all possible values that an attribute can take.

Example Relation

Student(StudentID, Name, Branch, Semester)

StudentID = Attribute

One student record = Tuple

Entire table = Relation

Keys in Relational Model

Keys are used to uniquely identify tuples and establish relationships between tables.

Primary Key

A primary key uniquely identifies each record in a relation and cannot contain duplicate or NULL values.

Candidate Key

A candidate key is an attribute or set of attributes that can uniquely identify a tuple.

Alternate Key

Candidate keys that are not selected as the primary key are called alternate keys.

Foreign Key

A foreign key is an attribute in one relation that refers to the primary key of another relation.

Referential Integrity

Referential integrity ensures that foreign key values always refer to valid primary key values in the related table.

Advantages of Relational Model

- Simple structure
- Easy data retrieval
- Data independence
- Strong integrity support
- Reduced redundancy

Applications

- Banking Systems
- E-Commerce Platforms
- Hospital Databases
- University Management Systems

Exam Important Questions

1. Define Relational Model.
2. Differentiate relation and tuple.
3. Explain primary key and foreign key.
4. What is referential integrity?
5. Explain different types of keys.

Summary

The Relational Model organizes data into tables and uses keys to establish relationships. It is the foundation of modern relational database systems such as MySQL, PostgreSQL, Oracle, and SQL Server.